

Pimlico: can we map individual bomb-damaged houses?

London Before | Research and executable pilot benchmark | 8 September 2026

Decision

Yes, we can build an index of historical houses and building parts, but the present overlay cannot support verified house-level damage claims. Keep the Three.js island renderer, improve the geometry unit, and build a separately evidenced historical index. A more detailed 3D map will not repair historical registration.

The pilot now shows only a Pimlico test crop, clipped to Westminster. One adjustable base colour covers the ground, slab edges and modern buildings; draft damage and cartographic detail can be toggled separately. The crop is explicitly a study extent, not an official neighbourhood boundary or the boundary of Sheet 88.

What the benchmark actually found

The saved OpenFreeMap snapshot contains 74 tile features inside the crop, but those features contain 1,053 disconnected polygon components. Forty-seven features contain multiple components; the largest bundles 388. Scoring each bundle as a single footprint diluted the damage overlap. The pilot now separates components while preserving courtyard holes and source-group references.

Before splitting, only 1 of 74 feature groups met the whole-footprint rule. After splitting, 34 of 1,053 polygon parts meet it, 55 have partial overlap and 964 have no overlap. All 20 draft areas intersect some modern geometry. These are different denominators: the increase is a correction to the matching unit, not a measured increase in historical accuracy. Unmatched does not mean undamaged.

The 1,053 parts share only 74 source feature IDs. The captured file retains no address fields. Therefore neither a source ID nor a polygon count can be presented as a house count. Local part IDs are reproducible within this saved snapshot and crop, not persistent property identifiers across recaptures.

Registration remains unmeasured

There are three approximate affine fitting anchors and zero independent check points. Their zero fitting residual is expected and does not establish accuracy. Deliberately translating the traces by 3 m changes 9-10 whole-footprint assignments; 5 m translations change 13-20. These are sensitivity experiments, not estimates of actual map error.

Reproduce with `pnpm benchmark:pimlico`. The generated `public/proto/london-island/benchmark.json` includes input hashes, capture provenance, scope, thresholds, eight translation experiments and a record for every part. Historical identity, damage classification and bomb-event accuracy are explicitly unscored.

Historical evidence: what can be established?

Keep damage, buildings and incidents separate

Sheet 88 is the LCC's 1945 cumulative damage record for Pimlico and Battersea, catalogue LCC_AR_TP_P_041_088, record 346262. Its colours describe damage to the wartime built environment. Repeated attacks and repairs can make the cause difficult to infer. It also includes V-weapon symbols, but a coloured polygon is not automatically an individual bomb impact. [London Picture Archive: Sheet 88](#).

The series comprises 110 hand-coloured maps at 1:2,500, on OS sheets originally published in 1916 and updated to 1940. Historical outlines should therefore be traced independently of current replacement buildings. Green and blue require context because clearance, open space and water can share colours. Map scale alone supplies no measured positional guarantee. [The London Archives: map explanation](#).

Sources to reconcile in the pilot

For dated incidents, The National Archives identifies HO 193 maps and tracings, HO 198 incident files and BC4 forms. These can connect dates, references and descriptions; incendiary incidents may cover areas rather than countable individual impacts. Individual Pimlico files were not inspected in this research. [TNA: Bomb Census guide](#).

Westminster's catalogue lists CD/22/142, a 1945 map of incidents during 1940-1945, with a copy in its search room. It distinguishes several weapon/incident types. This is a promising independent comparison, currently supported by catalogue evidence only; the underlying map has not been compared with our coordinates. [Westminster Archives: incident map](#).

Bomb Sight is useful for comparison, not a positional reference standard. Its main aggregate covers night bombing from 7 October 1940 to 6 June 1941. The creators document photography, georeferencing and manual digitising errors, without a numerical local error bound. Its published data terms restrict commercial reuse. [Bomb Sight: sources](#) and [copyright](#).

Historical addresses need naming and numbering evidence, directories and other property records. The Archives' guide identifies relevant street naming/numbering plans and house-history sources. Matching today's address string alone is insufficient. [The London Archives: house history](#).

The repository records Sheet 88 as ordered, with the delivered image and issued licence still pending. Verify the delivered resolution and permitted derived/vector reuse before replacing the present preview traces.

A small benchmark that can answer the house-level question

These are proposed project acceptance criteria, not an archival or surveying standard.

1. Freeze the evidence and register the original

Keep the delivered image unchanged, record its checksum and licence, and work in its native pixel coordinates. Establish approximately 12 distributed, unchanged ground features for fitting and at least 8 withheld check points. Avoid garden centres, imprecise bridge centres, changed corners and roof edges. Record how each point was identified, who reviewed it and its coordinate source.

Fit a simple affine transform first in British National Grid. Compare a more flexible transform only if held-out errors reveal local distortion. QGIS documents the control-point workflow and warns that higher-order transforms can distort areas away from control points. Preserve the control-point file and transformation settings. [QGIS 3.44: Georeferencer](#).

Report check-point median, RMSE, P95 and maximum error in metres, plus a spatial residual map. Never report training residuals as accuracy. Choose check points before tuning. The 20 points above are a practical starting budget, not a guarantee of sufficient spatial coverage.

2. Review 30 historical properties or parts

Select before tuning: 10 surviving terrace houses; 10 demolished or redeveloped properties; 5 mixed or partial-damage cases; and 5 ambiguous/unresolvable cases. This must start from historical records, so demolished houses are not silently excluded by sampling only today's buildings.

Cambridge Street, Warwick Square and St George's Square are candidate study locations. Russell House and the former Sutherland Terrace/Abbots Manor area supply useful rebuilding cases in the adopted plan. These are sampling anchors, not gold coordinates or pre-approved labels. [Westminster: adopted Pimlico plan, December 2022](#).

Two reviewers independently trace historical outlines and record colour, address evidence, modern relation and uncertainty. Adjudicate disagreements while retaining both initial judgements. Allow same building, replacement, split, merged, absent and unknown relationships. Ground truth must include explicit unresolvable cases.

3. Score separately, with an abstain option

Measure historical footprint IoU; damage-class agreement and per-class errors; correct property/address links; wrong-neighbour links; and abstention/coverage. Score dated incident linkage separately. Do not use the 60% overlap display rule as a calibrated probability.

Proposed first gate: held-out positional P95 below half the median frontage width measured in the sample; at least 90% correct property links among resolvable cases; all confidently wrong-neighbour links investigated; ambiguous records remain unassigned. Report raw counts and denominators. Thirty examples support a pilot decision, not a claim of London-wide accuracy.

Other maps and 3D models worth examining

Retain the current scene; compare data before switching renderers

The island uses Three.js; the earlier map uses MapLibre. Both can colour individual geometry. Our new common material colour already gives the ground and buildings the same base colour with natural shading. Higher roof detail, building identity and wartime identity are separate requirements.

Overture buildings: strongest open data comparison

Overture distributes buildings and building parts. Buildings have GERS IDs; parts reference a building through `building_id` and do not themselves receive GERS IDs. Source provenance helps distinguish OSM-derived data from other coverage. Heights and parts are optional, and geometries may represent roofprints. Compare a bounded extract against the reviewed sample; an OSM-derived Overture feature is not an independent accuracy check on OSM. [Overture: buildings](#) and [ODbL attribution](#).

Cesium OSM Buildings: selectable 3D precedent

Cesium supplies 3D Tiles with per-feature metadata and colour/hide support. Identity uses `elementType` together with `elementId`; heights may be estimated. It requires a Cesium ion account and attribution. This is a useful interaction reference, but shares OSM origins with the current map. [Cesium: OSM Buildings and live demo](#).

AccuCities and VU.CITY: detailed London geometry

AccuCities offers central London geometry and a free TQ3280 sample, which is not Pimlico. Its product claims 50 cm or 15 cm accuracy depending on level; these are vendor claims. The sample is for non-commercial use. Confirm the exact Pimlico delivery, date, separate building objects and public browser delivery rights before purchase. [AccuCities: model and samples](#).

VU.CITY sells editable geometry with separate terrain treatment and multiple export formats. It is a good visual reference for a physical neighbourhood model. Its public pages do not establish the exact Pimlico house IDs or the rights needed for our browser distribution. [VU.CITY: geometry and samples](#).

Google photorealistic tiles and Colouring Britain

Google's textured 3D mesh is a visual reference, but the reviewed evidence does not establish a reliable per-house index. Cesium engineering reported no per-building metadata in those tiles. Verify current capabilities and API policies before considering integration. [Google: Tile API](#) and [Cesium engineering response, April 2025](#).

Colouring Britain, the current successor to Colouring London, is an excellent reference for building records and attributes. Its downloadable data explicitly excludes footprint geometry for now, so it is not an available 3D geometry feed. [Colouring Britain: extracts](#).

The index and the next decision

An index we can maintain

Use our own immutable historical-site IDs. Each record needs a source sheet/reference, image polygon, registered geometry, historical address and valid dates, damage assessment, uncertainty and reviewer history. Keep source version and coordinate transformation alongside the record.

Store modern geometry in a separate versioned table with provider, source feature ID, building/part relationship, geometry and height provenance. Use a crosswalk for historical-to-modern relations: same, replacement, split, merged, absent or unresolved. Do not overwrite the old historical site when the modern map changes.

Store incidents separately with date or interval, weapon/type when evidenced, source reference, location geometry and uncertainty. A damage assessment can relate to several incidents; an incident can affect several properties. Keep unsupported links empty.

For surviving houses, combine footprint intersection, frontage orientation, street side, historical numbering and archival evidence. For redevelopment, show historical footprints as their own layer. Do not force them onto modern roofs. Historical heights need their own source or an explicit illustrative label.

What is ready now

The Pimlico crop, whole-model colour control, optional damage display, per-component inspection and repeatable snapshot benchmark are implemented. They let us find and discuss the exact modern geometry involved without asserting a wartime house identity.

The local computation now exposes the aggregation problem, measures sensitivity to small shifts and preserves the distinction between candidate and verified. Its polygon splitting restores a useful inspection unit but cannot recover party walls, missing addresses or geometry discarded upstream.

What should happen next

First, register the sharper Sheet 88 image and build the 30-property reviewed set. Second, compare a complete modern source extract, with building/part IDs and address evidence, against those cases. Overture is the first open comparison to try; retain OSM provenance so shared-source agreement is not double-counted. Third, decide whether basic extrusions suffice or a paid roof model adds enough value. Request licence and object-identity details before committing to a vendor.

A project-affiliated Layers of London research paper describes georeferenced/vectorised map work and a route for researchers to enquire about data. Investigate this before tracing every sheet manually, but no downloadable Pimlico layer, accuracy measurement or project-specific reuse permission was verified here. [Layers of London research paper, DOI 2024](#).

Evidence limits

Research combined two independent research lanes, primary archive/provider documentation, and local code/data inspection. Highest-impact archival and technical claims were spot-checked by the coordinating agent. Sources were accessed on 8 September 2026. TNA and Westminster catalogue access was intermittent; no incident file, commercial Pimlico model or Overture extract was measured. No claim of house-level accuracy is justified until the held-out and human-reviewed tests are completed. Further broad searches are unlikely to change that decision.